SONiX 32-Bit Cortex-M0 Micro-Controller

1M ADC Example Code

AMENDENT HISTORY

Version	Date	Description
1.0	2024/11/26	1. First version.

Table of Content

1 OVERVIEW

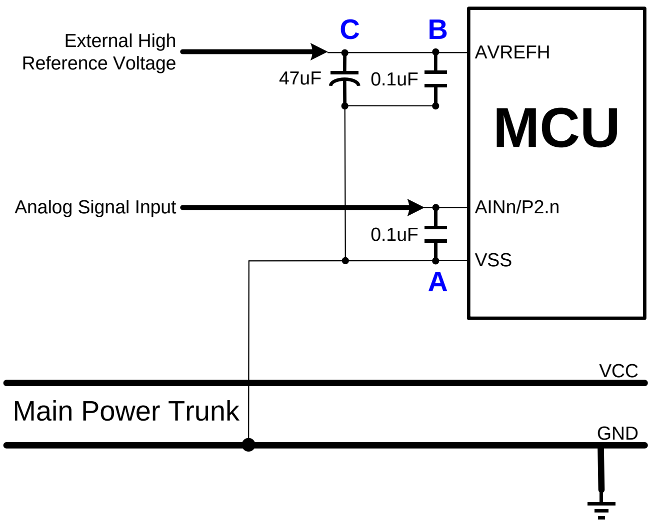
This document provides our customers the C program examples of ADC.

2 FEATURE

- ADC Type: Analog to Digital Convertor function.
- Sampling rate up to 1Msps
- Support auto-conversion mode
- Support ADB watchdog
- Use ARM-compiler V6.12

3 HW

- The starter kit board of vendor desired MCU.
- The ADC input signal must be through a 0.1uF capacitor. The 0.1uF capacitor is set between ADC input pin and VSS pin, and must be on the side of the ADC input pin as possible. Don't connect the capacitor's ground pin to ground plain directly, and must be through VSS pin. The capacitor can reduce the power noise effective coupled with the analog signal.



4 EXAMPLE CODE

4.1 ADC Initial

```c
/*****************************************************************************
* Function    : ADC_FuncInit
* Description : Initialization of ADC
* Input       : bPCLKDiv - ADC_DIV1, ADC_DIV2, ..., ADC_DIV32
*               bADCLen - ADC_8BIT, ADC_12BIT
*               bCHMode - Single_Channel, Multiple_Channel
*               bSCMode - Single_Mode, Continuous_Mode
* Output      : None
* Return      : None
* Note        : None
*****************************************************************************/
void ADC_FuncInit(uint8_t bPCLKDiv, uint8_t bADCLen, uint8_t bCHMode, uint8_t bSCMode)
{
  __ADC_ENABLE_HCLK;                                    //Enables HCLK for ADC

  SN_ADC->ADM_b.AVREFHSEL = ADC_AVREFHSEL_INTERNAL;    //Set ADC high reference voltage source from internal reference
  SN_ADC->ADM_b.VHS = ADC_VHS_INTERNAL_2V;            //Set ADC high reference voltage source as Internal 2V

  SN_ADC->ADM_b.GCHS = ADC_GCHS_EN;                     //Enable ADC global channel

  SN_ADC->ADM_b.ADLEN = bADCLen;                        //Set ADC resolution = 12-bit

  SN_ADC->ADM_b.ADCKS = bPCLKDiv;                       //ADC_CLK = ADC_PCLK/32

  SN_ADC->CONVCTRL_b.SCMODE = bSCMode;                  //Set mode

  SN_ADC->CONVCTRL_b.CH = bCHMode;                      //Set converting channel

  SN_ADC->ADM_b.ADENB = ADC_ADENB_EN;                   //Enable ADC

  UT_DelayNx10us(10);                                   //Delay 100us

  SN_ADC->ADM1_b.ACS = ENABLE;                          //Calibration start
  while (SN_ADC->ADM1_b.ACS == 1);

  SN_ADC->ADM1_b.CALIVALENB = ENABLE;                   //ADC conversion with calibration value
}
```

Call the function to set the ADC reference voltage source, ADC resolution, and ADC conversion time.

Example: Set ADC high reference voltage source as Internal 2V

Set ADC resolution as 12 bit

ADC conversion time = PCLK / 32

```
/*****************************************************************************
* Function    : ADC_InterruptInit
* Description : ADC Interrupt control of ADC channels
* Input       : bEOCALen - Enable bit refer to end of calibration
*               bOVRen - Enable bit refer to overrun
*               bAWWen - Enable bit refer to window watchdog
*               bEOSen - Enable bit refer to end of sequence
*               bADCCh - Enable bit refer to ADC channels
* Output      : None
* Return      : None
* Note        : None
*****************************************************************************/
void ADC_InterruptInit(uint8_t bEOCALen, uint8_t bOVRen, uint8_t bAWWen, uint8_t bEOSen, uint16_t bADCCh)
{
  SN_ADC->IE = ((bEOCALen << 27) | (bOVRen << 26) | (bAWWen << 25) | (bEOSen << 24) | (bADCCh));
  ADC_NvicEnable();                                      //Enable ADC NVIC interrupt
}
```

Call the function to control over which A/D channels generate an interrupt when a conversion is complete.

4.2 ADC Example

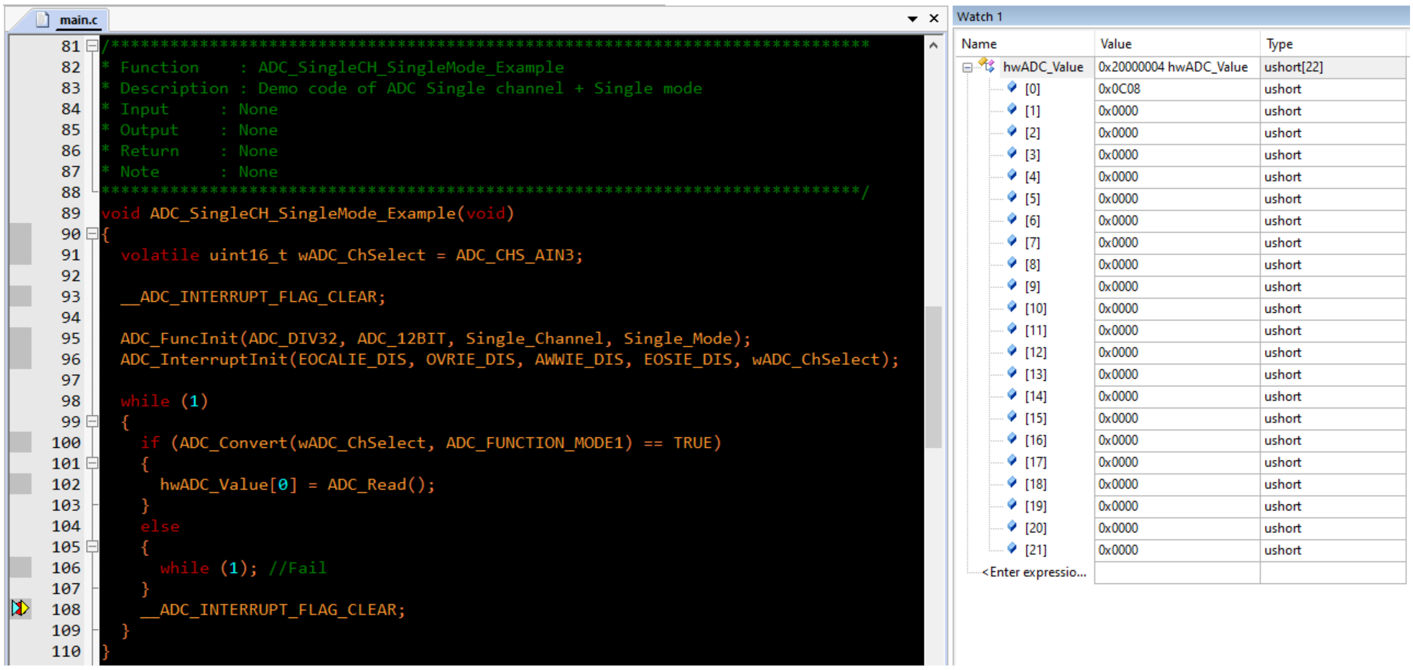
```
  ADC_SingleCH_SingleMode_Example();
  ADC_SingleCH_ContinuousMode_Example();
  ADC_MultipleCH_SingleMode_Example();
  ADC_MultipleCH_ContinuousMode_Example();
  ADC_Window_Watchdog_Example();
```

There are 5 types of examples: single-channel single mode, single-channel continuous mode, multi-channel single mode, multi-channel continuous mode and ADB watchdog example.

5 RESULT

5.1 ADC Type

The ADC converter result can be observed from the Watch window.



main.c

```c
/*****************************************************************************
* Function     : ADC_SingleCH_SingleMode_Example
* Description : Demo code of ADC Single channel + Single mode
* Input       : None
* Output      : None
* Return      : None
* Note        : None
*****************************************************************************/
void ADC_SingleCH_SingleMode_Example(void)
{
  volatile uint16_t wADC_ChSelect = ADC_CHS_AIN3;

  __ADC_INTERRUPT_FLAG_CLEAR;

  ADC_FuncInit(ADC_DIV32, ADC_12BIT, Single_Channel, Single_Mode);
  ADC_InterruptInit(EOCALIE_DIS, OVRIE_DIS, AWWIE_DIS, EOSIE_DIS, wADC_ChSelect);

  while (1)
  {
    if (ADC_Convert(wADC_ChSelect, ADC_FUNCTION_MODE1) == TRUE)
    {
      hwADC_Value[0] = ADC_Read();
    }
    else
    {
      while (1); //Fail
    }
    __ADC_INTERRUPT_FLAG_CLEAR;
  }
}
```

Watch 1

Name	Value	Type
hwADC_Value	0x20000004 hwADC_Value	ushort[22]
[0]	0x0C08	ushort
[1]	0x0000	ushort
[2]	0x0000	ushort
[3]	0x0000	ushort
[4]	0x0000	ushort
[5]	0x0000	ushort
[6]	0x0000	ushort
[7]	0x0000	ushort
[8]	0x0000	ushort
[9]	0x0000	ushort
[10]	0x0000	ushort
[11]	0x0000	ushort
[12]	0x0000	ushort
[13]	0x0000	ushort
[14]	0x0000	ushort
[15]	0x0000	ushort
[16]	0x0000	ushort
[17]	0x0000	ushort
[18]	0x0000	ushort
[19]	0x0000	ushort
[20]	0x0000	ushort
[21]	0x0000	ushort
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